

[INITIATING PLAYER_01 & PLAYER_02]

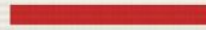
OPTIMASI KEUNTUNGAN MENGUNAKAN METODE GRAFIS

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TEKNIK RISET OPERASI

COMPUTER SCIENCE DEPARTMENT

STAGE 01



ANALISIS & LATAR BELAKANG

IDENTIFIKASI SUMBER DAYA TERBATAS

MASALAH SUMBER DAYA

Stok Bahan Baku

Kapasitas susu harian dibatasi maksimal 10.000 ml. Membatasi produksi Menu X1 dan X2 secara simultan.

Alokasi Barista

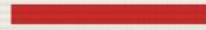
Waktu kerja efektif barista hanya 400 menit per hari. Setiap menit dihitung untuk efisiensi total.

OBJECTIVE: Menentukan titik produksi (X1, X2) untuk profit maksimal.

RUMUSAN MASALAH UTAMA

-  Formulasi Model: Bagaimana menyusun kendala operasional ke dalam model matematika Linear Programming?
-  Kombinasi Optimal: Berapa kuantitas cup Kopi Susu dan Matcha Latte yang harus diproduksi?
-  Alur Komputasi: Bagaimana merancang program Python untuk otomatisasi visualisasi Feasible Zone?

STAGE 02



MODEL & KALKULASI

MATHEMATICAL ENGINE PROCESSING

FORMULASI MODEL MATEMATIKA

Parameter	Model Persamaan / Pertidaksamaan
Fungsi Tujuan	$\text{Max } Z = 15000x_1 + 20000x_2$
Kendala Susu	$100X1 + 200X2 \leq 10.000$
Kendala Waktu	$5X1 + 5X2 \leq 400$
Non-Negatif	$X1, X2 \geq 0$

KALKULASI MANUAL SUMBU

Garis Kendala 1

$X_1=0 \rightarrow X_2=50$ (A: 0, 50)

$X_2=0 \rightarrow X_1=100$ (B: 100, 0)

Garis Kendala 2

$X_1=0 \rightarrow X_2=80$ (C: 0, 80)

$X_2=0 \rightarrow X_1=80$ (D: 80, 0)

ELIMINASI: $X_1 + 2X_2 = 100$ | $X_1 + X_2 = 80$ ► SOLUSI: TITIK E (60, 20)

| SOLUSI OPTIMUM TERCAPAI

1.3M

RUPIAH PROFIT / HARI

TARGET PRODUKSI:

- ▶ 60 CUP KOPI SUSU (X1)
- ▶ 20 CUP MATCHA LATTE (X2)

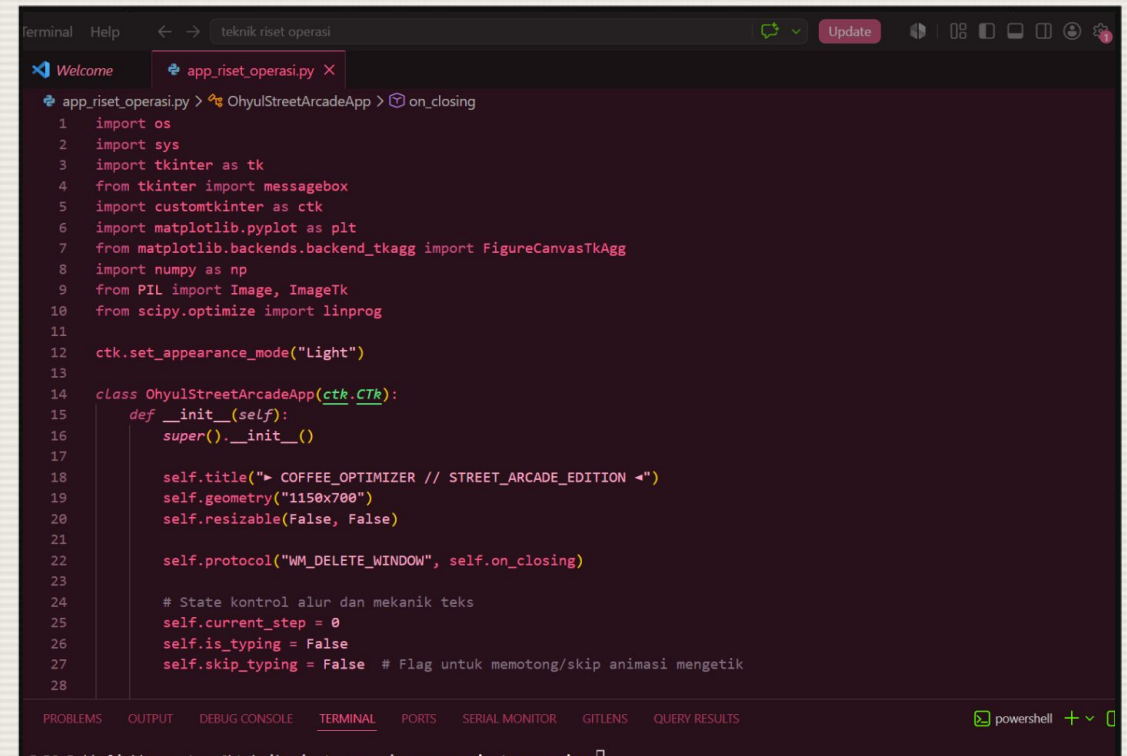
Resource Slack = 0 (Maximum Efficiency)

DESKRIPSI SOFTWARE ENGINE

Tech Stack Matrix

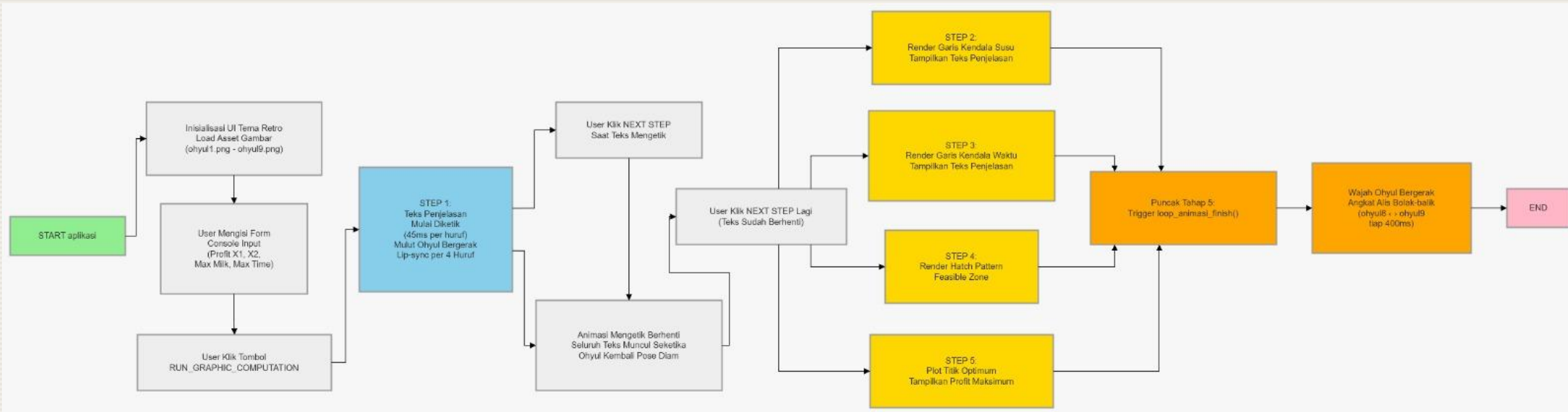
Program dikembangkan menggunakan Python 3.13 dengan integrasi library tingkat lanjut untuk kalkulasi linear.

- SciPy: Algoritma optimasi Highs.
- Matplotlib: Visual engine retro hatching.
- CustomTkinter: Arcade GUI interface.
- Pillow: Image processing Ohyul avatar.

A screenshot of a Visual Studio Code editor window. The title bar shows 'terminal Help' and a search bar with 'teknik riset operasi'. The editor has two tabs: 'Welcome' and 'app_riset_operasi.py'. The code in the file is a Python script for a game engine. It starts with imports for os, sys, tkinter, messagebox, customtkinter, matplotlib, numpy, PIL, and scipy. It then sets the appearance mode to 'Light'. A class 'OhyulStreetArcadeApp' is defined, inheriting from 'CustomTkinter'. The class has an '__init__' method that sets the title to 'COFFEE_OPTIMIZER // STREET_ARCADE_EDITION', geometry to '1150x700', and makes the window non-resizable. It also sets a protocol for the window. The class has attributes for 'current_step', 'is_typing', and 'skip_typing'.

```
1 import os
2 import sys
3 import tkinter as tk
4 from tkinter import messagebox
5 import customtkinter as ctk
6 import matplotlib.pyplot as plt
7 from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg
8 import numpy as np
9 from PIL import Image, ImageTk
10 from scipy.optimize import linprog
11
12 ctk.set_appearance_mode("Light")
13
14 class OhyulStreetArcadeApp(ctk.CTk):
15     def __init__(self):
16         super().__init__()
17
18         self.title("> COFFEE_OPTIMIZER // STREET_ARCADE_EDITION <")
19         self.geometry("1150x700")
20         self.resizable(False, False)
21
22         self.protocol("WM_DELETE_WINDOW", self.on_closing)
23
24         # State kontrol alur dan mekanik teks
25         self.current_step = 0
26         self.is_typing = False
27         self.skip_typing = False # Flag untuk memotong/skip animasi mengetik
28
```

ALUR PROGRAM



PROGRAM DEMO

READY TO RUN?

<https://github.com/hafsahha/Ohyu1-Street-Arcade>

► STARTING DEMO SESSION

THANK YOU



GAME CLEAR // MISSION SUCCESS

ICA & ACA

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